

Assignment-3.5

“Case Study Practical”

TASK 1. A user reports their laptop won't power on even when plugged in. List 5 troubleshooting steps you would take to diagnose the cause, and for each step, explain what you are checking for.

Step 1: Check the Power Outlet and Charger LED

Plug the charger into a known working wall outlet and check if the status light on the power brick turns on. You are checking whether electricity is coming from the wall and if the charger block is functional.

Step 2: Inspect the Charging Cable and Laptop DC Jack

Examine the charging cord for physical tears, sharp bends, or exposed wires, and look inside the laptop's charging port with a flashlight. You are checking for damaged wires, bent center pins, or trapped dust that blocks a solid physical connection.

Step 3: Perform a Hard Static Reset

Unplug the charger, disconnect the battery if removable, and hold down the laptop power button for 30 seconds before reconnecting the charger. You are checking if static electricity trapped in the motherboard capacitors is preventing the system from starting up.

Step 4: Test the Battery Bypass Method

Remove the laptop's internal or external battery completely, connect only the power adapter, and press the power button. You are checking if a shorted or completely dead battery is blocking power from reaching the rest of the laptop.

Step 5: Test the Charger Voltage Output with a Multimeter

Set a multimeter to DC voltage, place the red probe inside the tip of the charger barrel, and place the black probe on the outside metal ring. You are checking whether the charger is delivering the exact DC voltage required on its product label.

TASK 2. You receive a help desk ticket: 'My monitor is showing a blank screen, but the CPU seems to be running.' Write a step-by-step troubleshooting flow you would follow to identify and resolve the issue. Consider checking both hardware connections and basic display settings.

Step 1: Check Monitor Power

Check if the power LED light on the monitor is turned on. If the light is off, reconnect the power cable or test the wall socket.

Step 2: Check Cable Connection

Ensure the HDMI or DisplayPort cable is plugged firmly into the dedicated graphics card port at the bottom back of the PC case, not the motherboard port at the top.

Step 3: Reset Screen Settings

Press **Windows Key + Ctrl + Shift + B** to reset the Windows graphics driver, or press **Windows Key + P** to select the primary display.

Step 4: Test Monitor on Another Computer

Connect the monitor to a working laptop or second system to confirm whether the monitor or cable itself is defective.

Step 5: Reseat the RAM Modules

If the monitor works on another system, turn off the PC, open the case, and re-insert the RAM sticks into their slots. Loose RAM allows fans to spin but prevents any display output.

TASK 3. A user complains that their wireless mouse is unresponsive, but the keyboard works fine. Describe how you would isolate whether the issue is with the mouse, the USB port, or the system settings.

1. Isolate the Mouse (Hardware Check):

Turn the mouse upside down and verify that the power switch is turned ON and the optical LED sensor lights up. Replace the internal AA or AAA battery with a fresh one, turn the switch off and on, and press the dedicated sync button to re-pair the mouse with its USB dongle.

2. Isolate the USB Port (Port Check):

Unplug the wireless USB receiver dongle from its current USB port and plug it into a different port directly on the motherboard (at the back of the PC case). Since the keyboard already works fine in its own port, plug the mouse dongle into the exact same USB port where the keyboard was previously plugged in. If the mouse starts working, the original USB port is damaged or disabled.

3. Isolate System Settings (Software Check):

Use the keyboard to press the **Windows Key**, type **Device Manager**, and navigate using the arrow and **Tab** keys to open it. Locate the **Mice and other pointing devices** section, select your mouse driver, press **Shift + F10** to open the context menu, and select **Uninstall device**. Unplug the USB receiver, plug it back in, and allow Windows to automatically reinstall the default mouse driver.

TASK 4. Imagine you are working on a helpdesk team for a gaming cafe. One system is overheating and shutting down during use. List 3 possible hardware causes and describe how you would check each one. Think about fans, dust, and thermal paste.

1. Dust Accumulation in Heat Sinks and Ventilation Vents

How to Check: Open the computer case and visually inspect the CPU cooler, graphics card heat sink fins, and intake case fans. Heavy dust buildup acts like a blanket, trapping heat inside the cooling fins and blocking cold airflow across critical components.

2. Malfunctioning or Dead Cooling Fans

- **How to Check:** Power on the PC while the side panel is open and observe all fans. Check if the CPU cooling fan, GPU fans, and case fans are spinning smoothly without making grinding noises. Place your hand near the exhaust vents to feel if air is actively moving out of the system.

3. Dried or Improperly Applied Thermal Paste

- **How to Check:** Remove the CPU cooler from the processor socket and examine the thermal paste layer on top of the CPU heat spreader. If the paste is dry, cracked, flaky, or missing entirely, it can no longer transfer heat efficiently to the cooler. Clean off the old paste with isopropyl alcohol, apply a fresh pea-sized dot of thermal paste, and remount the cooler securely.